

Session 7 — The Sun, safely

Practical Astronomy Intensive

7 questions 20 minutes 3 concepts 5 diagnostic

SAFETY GATE, not a scored paper. The first three questions are pass/fail: a student who fits an eyepiece filter, trusts smoked glass, or thinks a quick glance is survivable does not touch a telescope in daylight until retaught and rechecked. Mark these before the Sun is up.

- 1 Two solar filters are on the table: one screws into the eyepiece barrel, one caps the front of the tube. Which do you fit, and why?

solar-observing Recall Multiple choice

- a The eyepiece filter — it sits closest to the eye, which is the last line of defence
REVEALS DANGER: eyepiece-end solar filter believed safe — the focused Sun cracks it with the eye right behind
- b The eyepiece filter, with the front left open so you can still aim the scope
REVEALS DANGER: eyepiece filter plus unfiltered aperture — the worst configuration there is
- c** The front-aperture filter, always — it rejects the sunlight before the telescope concentrates it, so nothing downstream ever carries the heat ✓
- d Either; both are sold as solar filters, so both are certified for the job
REVEALS DANGER: filter position treated as interchangeable — an eyepiece filter is not a safe substitute

An eyepiece-end filter sits at the focus of the entire aperture: the telescope concentrates the Sun's full heat onto that scrap of glass, and it can crack without warning with the observer's eye directly behind it. This is a known killer of eyes, not a theoretical concern. Filters go over the front. Always. And the finder gets capped or filtered too — an uncapped finder will blind you just as fast as the main tube.

- 2 Someone offers you these for looking at the Sun: dark sunglasses, a strip of exposed photographic film, a piece of smoked glass, and a welder's glass of shade 10. The Sun looks comfortably dim through every one. Which is safe?

solar-observing Apply Multiple choice

- a** None of them — all four cut the visible glare while letting infrared through, and comfort is not the test. Only a certified solar filter (ND5 front-aperture film, or shade 14+ welding glass) is safe ✓
- b The shade 10 welding glass — welders look at an arc through it all day
REVEALS DANGER: welding glass below shade 14 believed adequate for the Sun
- c The exposed film and the smoked glass — these are the traditional methods and have been used for generations
REVEALS DANGER: film and smoked glass believed safe — they pass infrared and burn the retina
- d The sunglasses, if you stack two pairs so the Sun looks properly dim
REVEALS DANGER: apparent dimness treated as the safety test — sunglasses pass infrared at any stacking

The whole trap is that all four LOOK fine. They dim the visible band, which is what your comfort responds to, and pass the infrared that does the burning. Worse, the retina has no pain nerves: the damage arrives with no sensation at all, and the blind spot shows up hours later, permanently. Comfort is not evidence of safety — certification is.

- 3 Set up the refractor for the Sun. Fit the front-aperture filter, deal with the finder, and hand the scope to the instructor for check — do not put your eye, or let anyone else put an eye, to the eyepiece until the instructor has personally passed it.

solar-observing

Apply

Practical task

ANSWER / WHAT TO WATCH FOR

The instructor personally verifies, every time, before any eye goes near the eyepiece — this check is not delegated to another student and is not skipped for a student who has done it before. Watch for and require: (1) front-aperture filter seated square and FULLY covering the aperture, held so it cannot be lifted off by wind or a knock — set screws or tape, not friction alone; (2) filter film inspected against the sky first for pinholes, scratches or creases — any defect and the filter is rejected outright, not patched; (3) the finder capped or filtered — an uncapped finder blinds as fast as the main tube; (4) NO eyepiece-end solar filter present anywhere on the setup; (5) the scope aimed by shadow, not by sighting along the tube; (6) the student waits for the instructor's word before the first look, and does not step away leaving an aimed unattended scope. Any student who fitted an eyepiece filter, left the finder open, or looked before the check does not observe the Sun today. Reteach, recheck, and only then hand it back.

This is the one check in the camp that is pass/fail with no partial credit, and the reason it is a practical and not an MCQ is that a student can recite the rule and still leave the finder open. The instructor's eyes on the fitted filter, every single time, is the protocol.

- 4 Through the filtered scope, a sunspot group shows as a dark blotch on the disc. What is a sunspot?

solar-activity

Apply

Multiple choice

- a A shadow cast on the photosphere by a prominence looping above it
REVEALS Sunspots taken as shadows rather than cooler magnetic regions
- b A hole in the Sun's surface, through which you are seeing down into a darker interior
REVEALS Sunspots believed to be holes in the Sun
- c A region where a strong magnetic field chokes convection, leaving the gas at around 4,000 K — cool only beside the 5,800 K photosphere around it; lifted out on its own it would blaze brighter than an arc lamp
✓
- d A genuinely cold, dark patch of the Sun — cold in the ordinary sense of the word
REVEALS Sunspots believed absolutely cold — a spot is ~4,000 K and would blaze if seen alone

'Dark' here is purely relative — the spot is darker than its surroundings by contrast, and at 4,000 K it is still radiating fiercely. Nothing is missing from the Sun where a spot sits; the magnetic field is just holding the hot convection back.

- 5 We call the photosphere the Sun's 'surface' and it looks like a crisp edge through the filter. What is it actually?

solar-structure

Apply

Multiple choice

- a A shell of denser gas held up by the pressure of the corona pressing down from outside
REVEALS Sun's structure inverted — the corona is thin and hot, not a compressing lid
- b Just the depth at which the gas stops being opaque and light finally escapes — there is no boundary there, only a change in transparency ✓
- c A solid or liquid surface, with the atmosphere sitting above it as on Earth
REVEALS Sun believed to have a solid surface
- d The top of the fusion core, where the energy is actually generated
REVEALS Photosphere confused with the core — fusion happens at 15 million K, far below

The edge looks sharp because the transition from opaque to transparent happens over a few hundred km out of 700,000 — sharp on the eye, gradual in the physics. The mottled granulation you can see in it is the top of the convection zone boiling, like a pot of oatmeal.

- 6 With the checked filtered scope, find the sunspot groups on today's disc, sketch their positions on the blank disc template, and count the groups. Then say — out loud — whether today's count says anything about where we are in the solar cycle.

solar-activity

Apply

Practical task

ANSWER / WHAT TO WATCH FOR

Watch for: a disc sketch with spots placed at plausible latitudes (spots cluster within $\sim 30^\circ$ of the equator, not at the poles); umbra and penumbra distinguished in at least one group; a group count stated as a count of GROUPS, not individual spots. On the verbal check they should recognise that one day's count says little on its own — the 11-year cycle shows up only in counts pooled over months, which is exactly what makes sunspot counting a citizen-science programme rather than a one-night observation. Push back if they describe the spots as holes or as cold, and if they say activity is a remote curiosity, ask what drives the aurorae and what a geomagnetic storm does to GPS.

Sunspot sketching is the oldest continuous dataset in astronomy and an amateur can still add to it. It also makes the relative-darkness point physically: the spot is obviously part of the same glowing surface, just dimmer.

- 7 The core is 15 million K. The photosphere is 5,800 K. What does the temperature do in the corona above it, and why is that interesting?

solar-structure

Reason

Multiple choice

- a It holds steady at photosphere temperature, since the corona is just thinner gas at the same heat

REVEALS Corona assumed thermally continuous with the photosphere

- b It drops to near absolute zero — the corona is essentially vacuum

REVEALS Low density confused with low temperature

- c It keeps falling smoothly outwards, as you would expect moving away from the heat source

REVEALS Sun assumed to cool steadily outwards — the corona is far hotter than the photosphere below it

- d It jumps back up to millions of kelvin — hundreds of times hotter than the surface beneath it, which nobody has fully explained ✓

The coronal heating problem is a live research question: the gas gets hotter as you move AWAY from the source of energy, which no simple thermal picture allows. Magnetic reconnection and wave heating are the leading candidates and neither fully closes it. Worth saying out loud — students should meet an unsolved problem in the middle of a settled subject.